

COMP 3411/9414

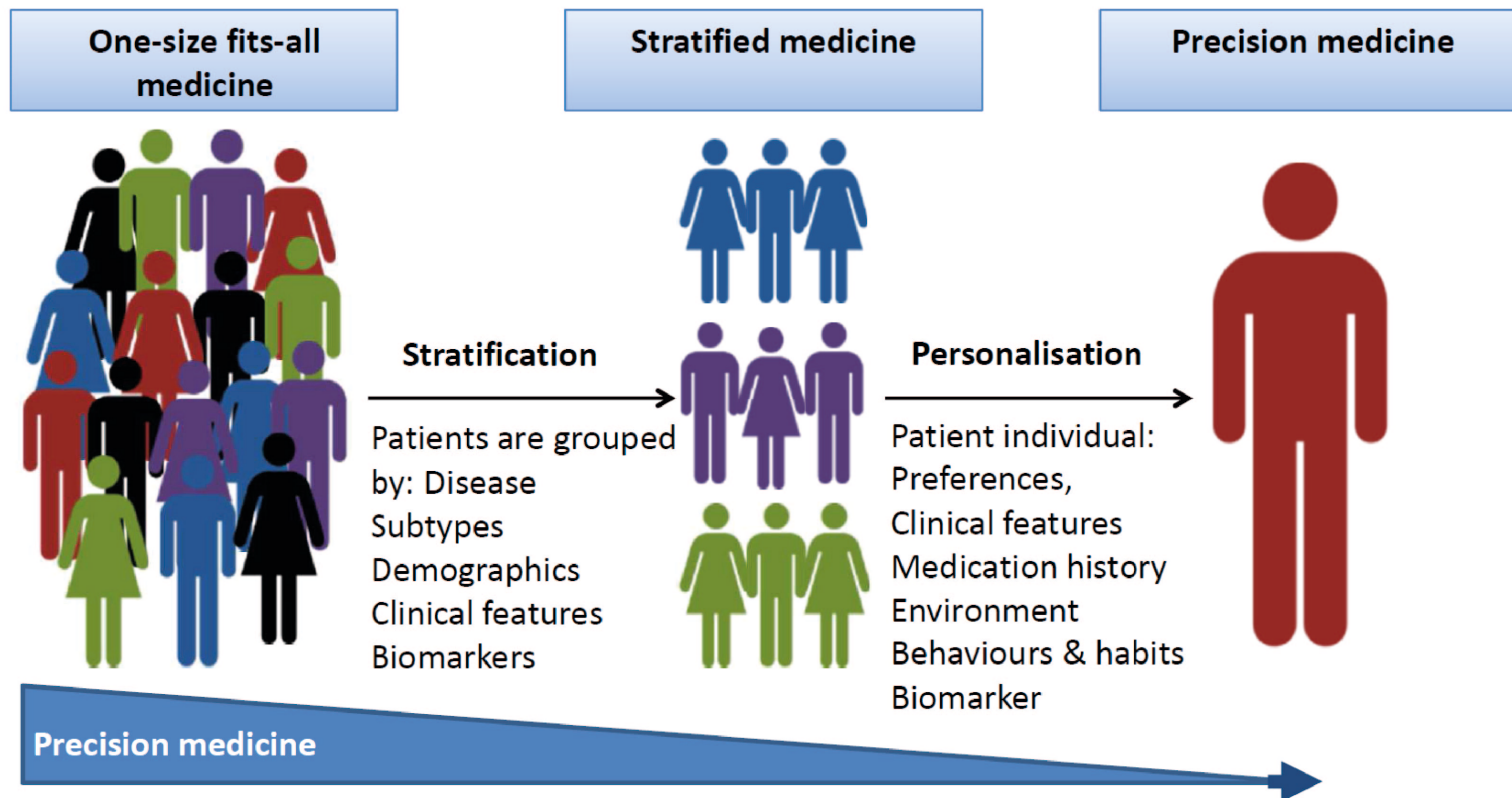
Artificial Intelligence

AI in Computer Vision – Part 3

AI in Biomedical Imaging

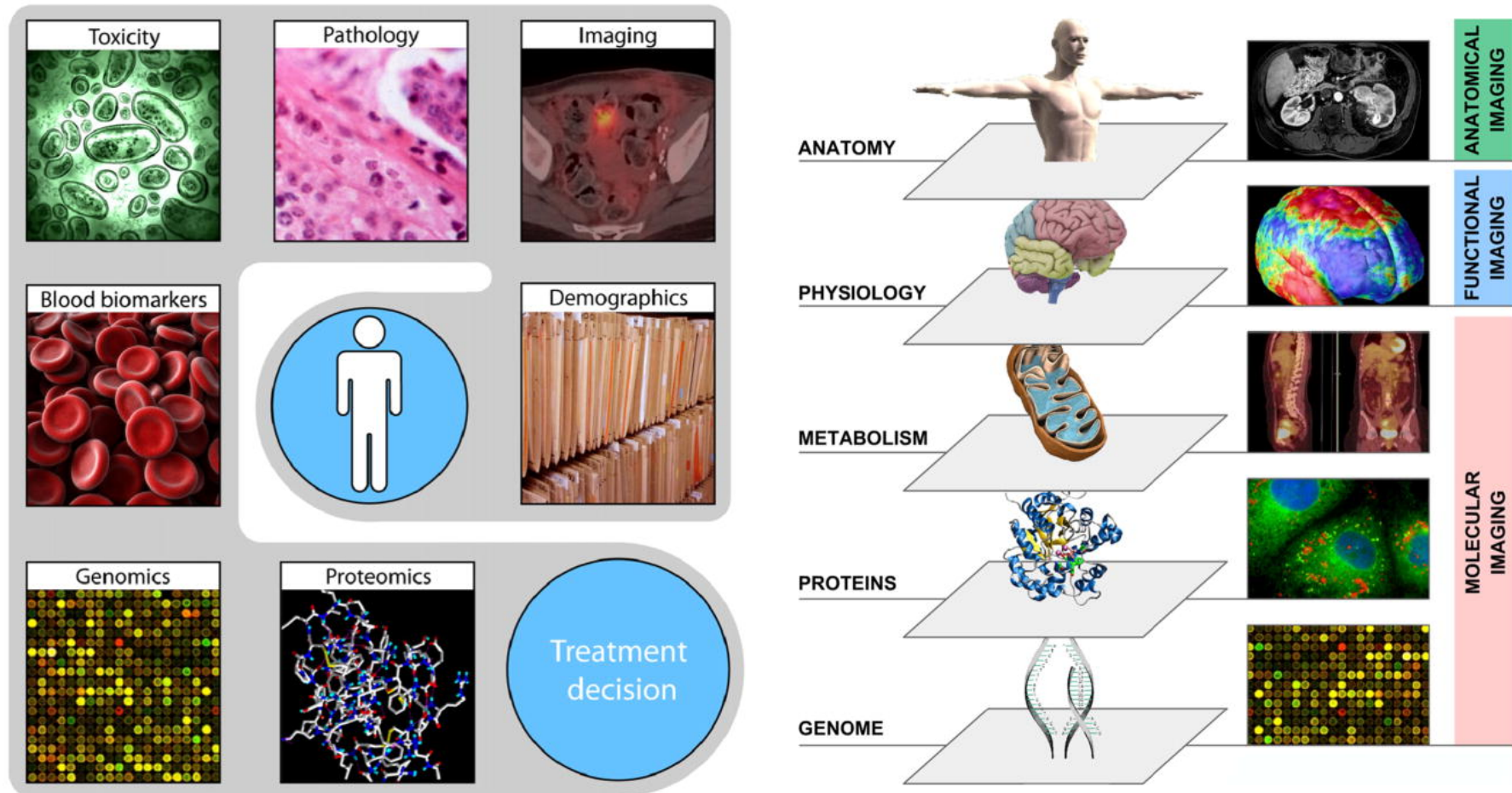
- Why do we need Biomedical Images?
- Why do we need AI for it?
- Is there any particular challenge in this?
- How do we address these challenges?

Precision Medicine



Source: Manchester Precision Medicine Institute

Biomedical Imaging in Precision Medicine

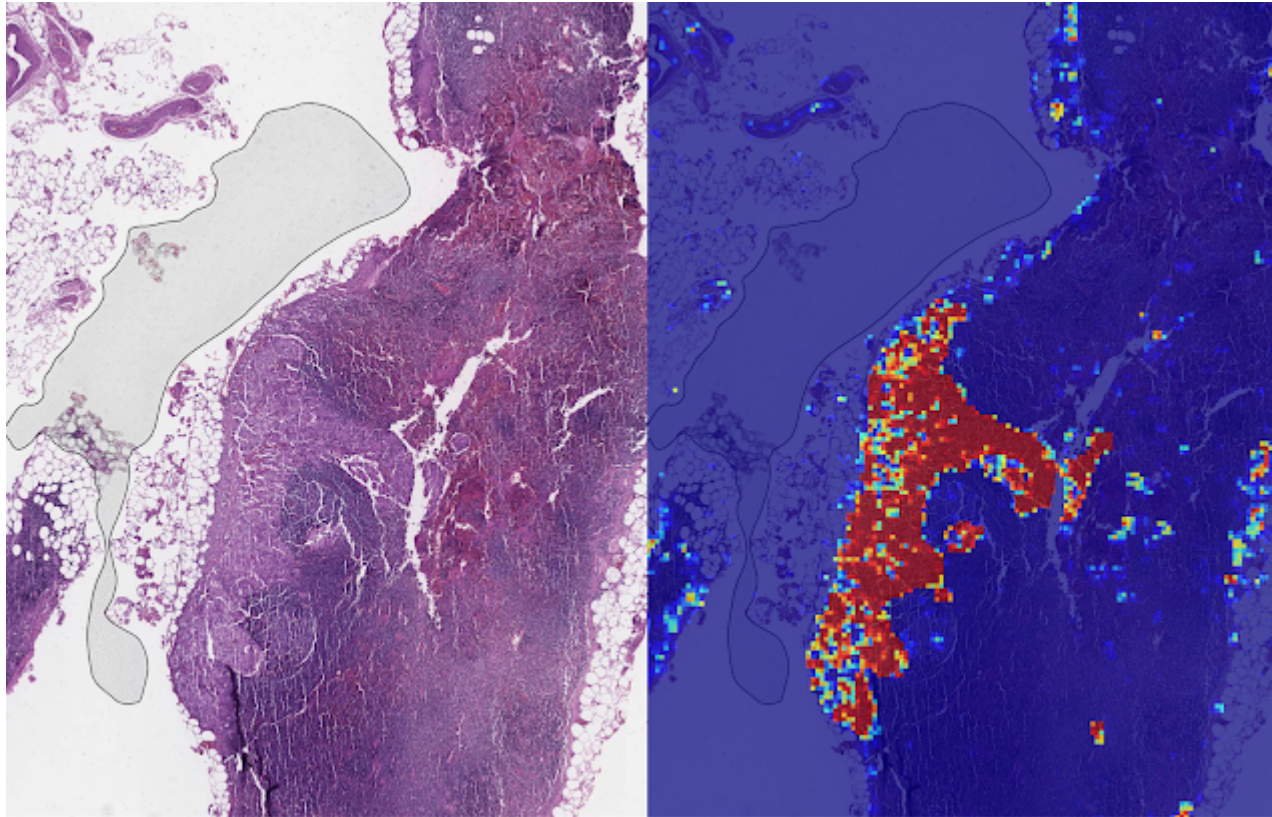


Source: P. Lambin *et al.*, *Eur J Cancer* 48(4):441-446, 2012

AI in Biomedical Imaging

- Biomedical services and research produce lots of imaging data
- With big data, analysis becomes a huge problem
- AI is essential in achieving automated, effective and reproducible analysis of biomedical imaging data
- There are however many unique challenges with biomedical images compared to general non-medical images
- Companies such as GE, Siemens and Google invest heavily in AI-based biomedical imaging applications

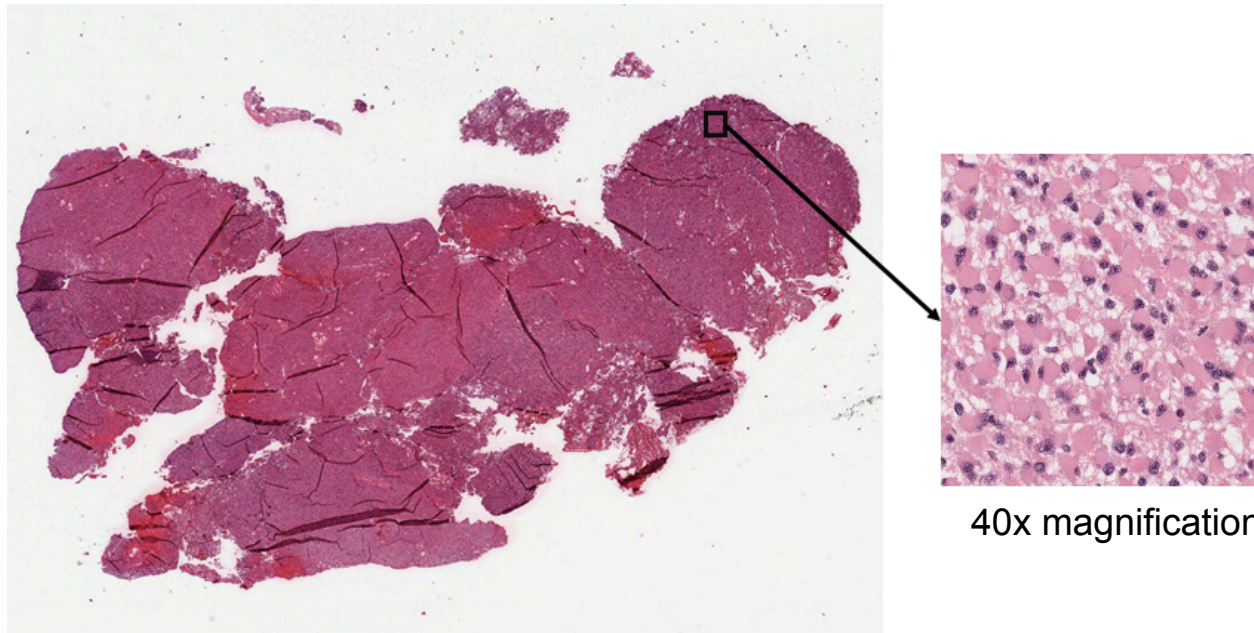
Cancer Detection



Source: <https://ai.googleblog.com/2018/10/applying-deep-learning-to-metastatic.html>

Cancer Detection

- Whole-slide image (WSI): very high resolution images
- A shift to fully digital environment for pathology



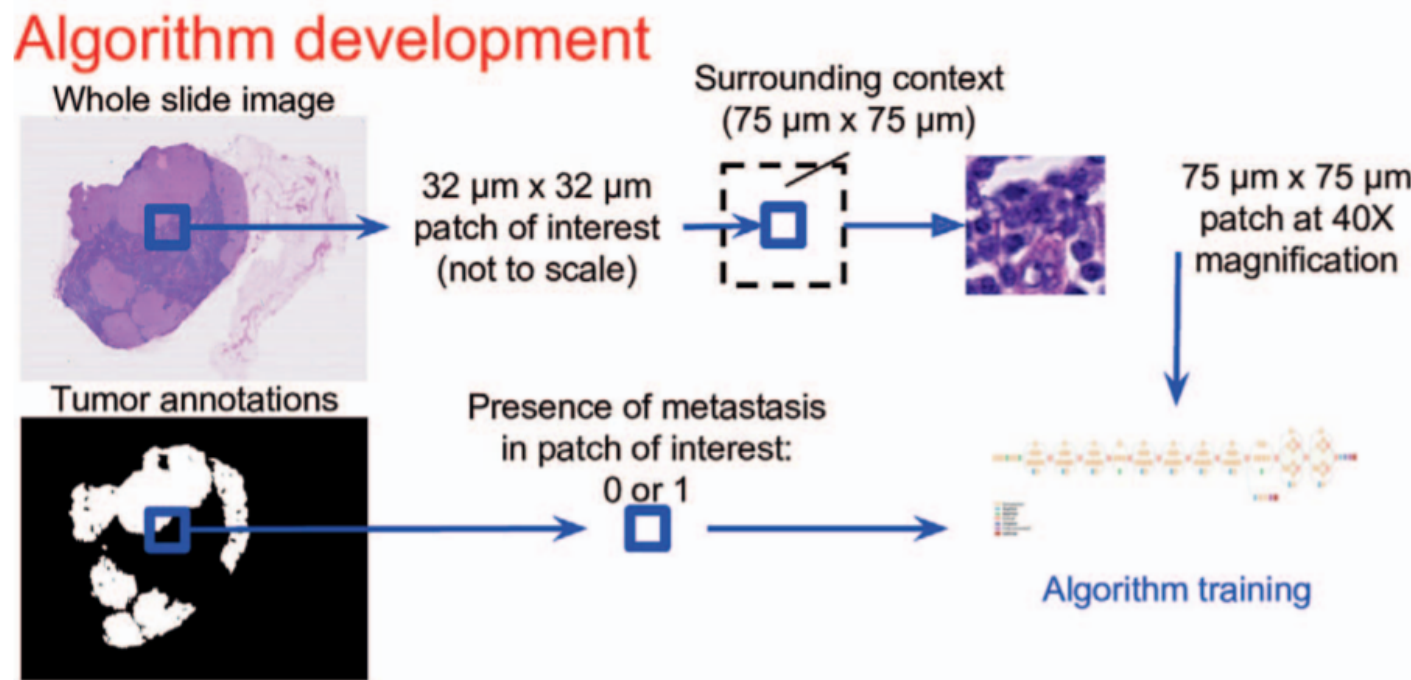
Cancer Detection



Source: <https://ai.googleblog.com/2018/10/applying-deep-learning-to-metastatic.html>

Cancer Detection

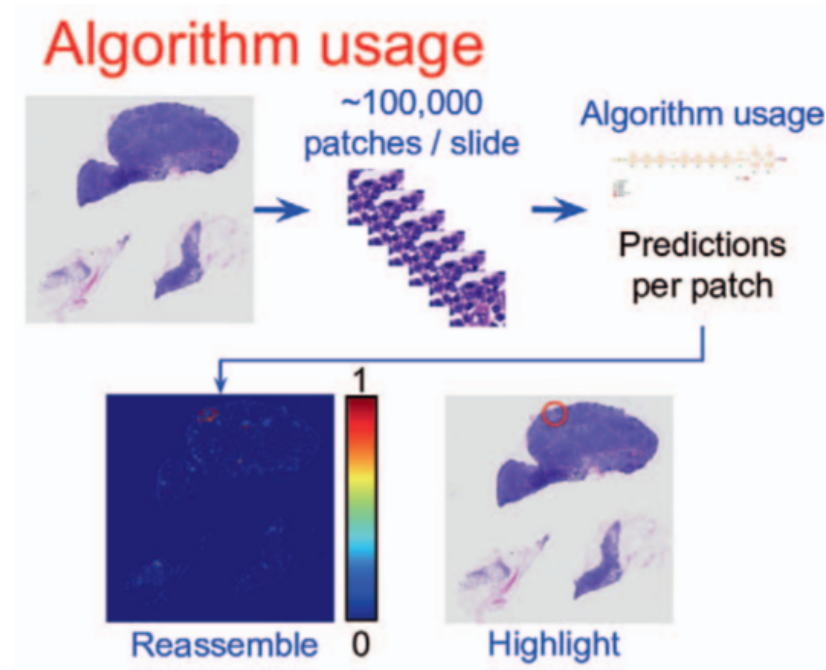
- Training stage:
 - Patch-wise processing with patch-level labels



Source: Y. Liu et al. *Artificial intelligence-based breast cancer nodal metastasis detection*. Arch Pathol Lab Med, 2018.

Cancer Detection

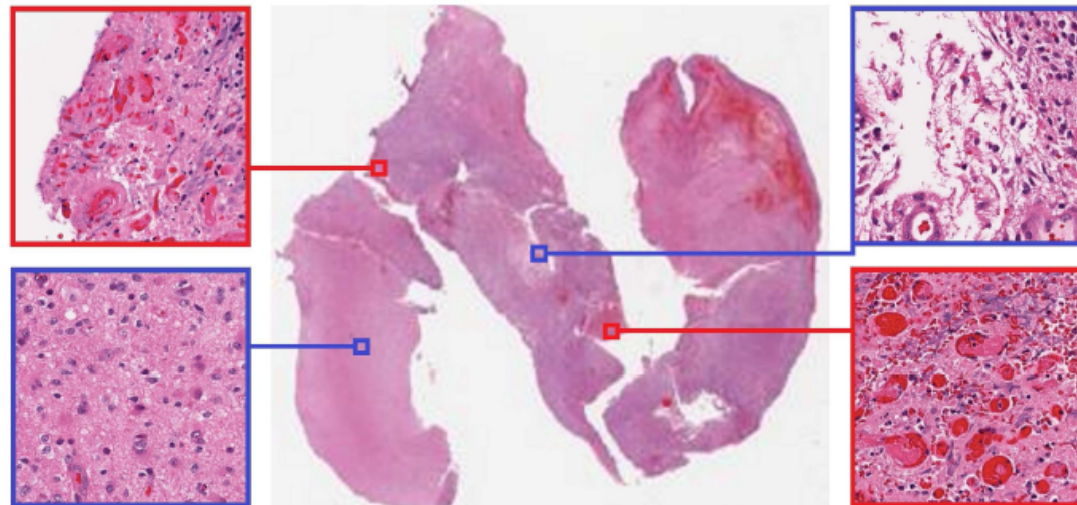
- Testing stage:
 - Patch-wise classification



Source: Y. Liu et al. *Artificial intelligence-based breast cancer nodal metastasis detection*. Arch Pathol Lab Med, 2018.

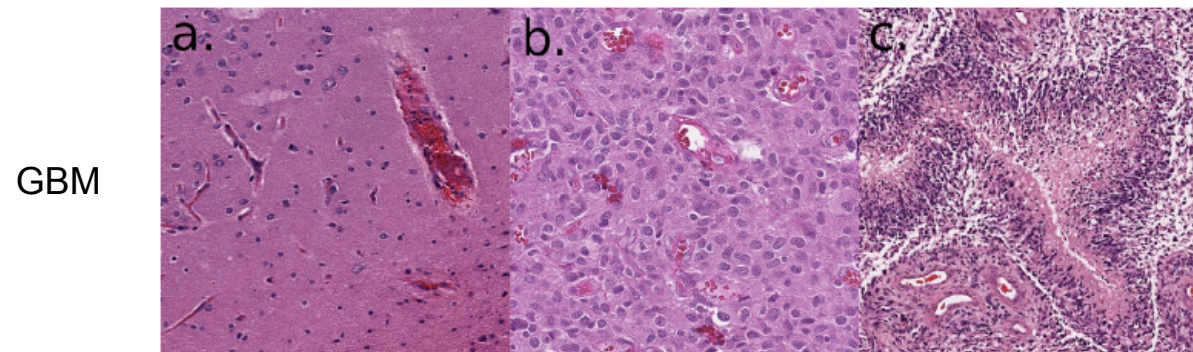
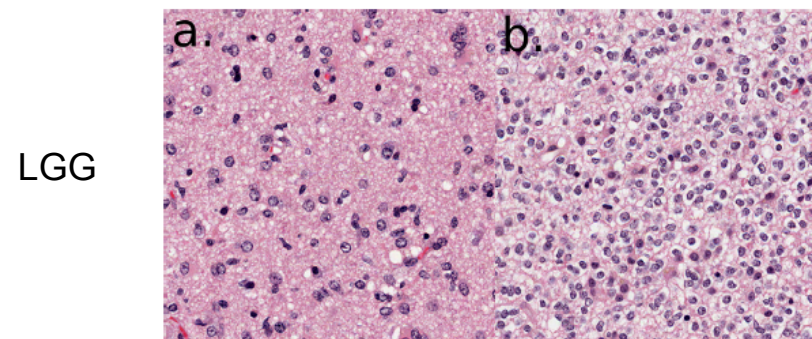
Real Challenges

- Challenge I:
 - Large image with image-level label only



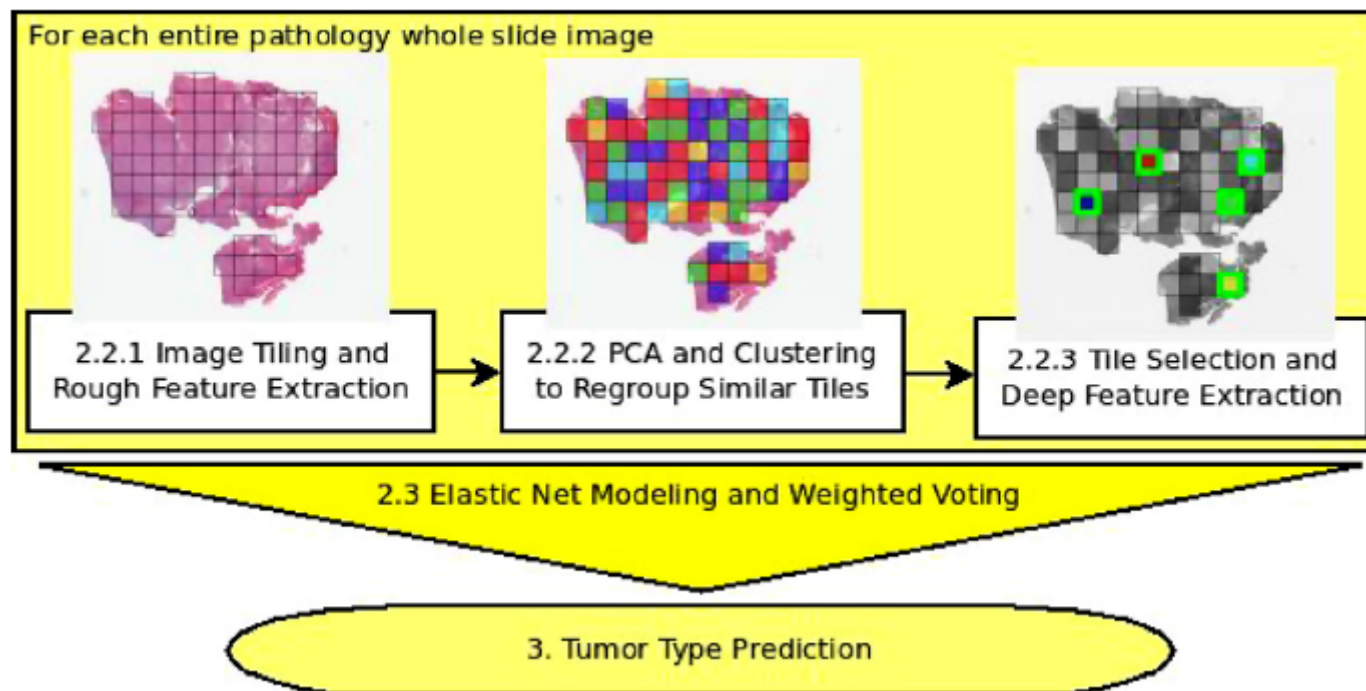
Real Challenges

- Challenge II:
 - Histology heterogeneity (subtypes and regional variations)



Clustering-based

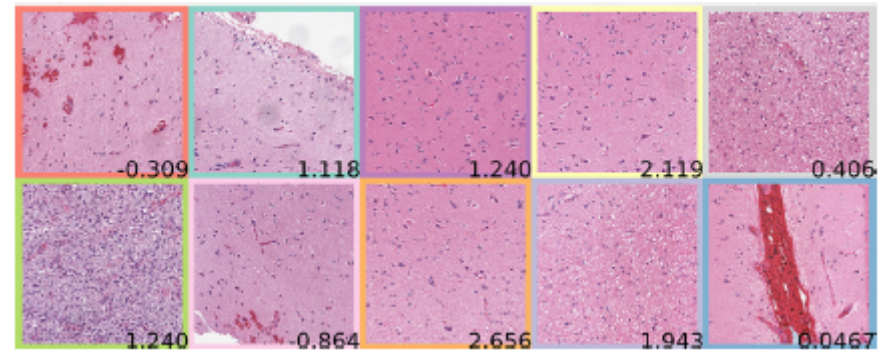
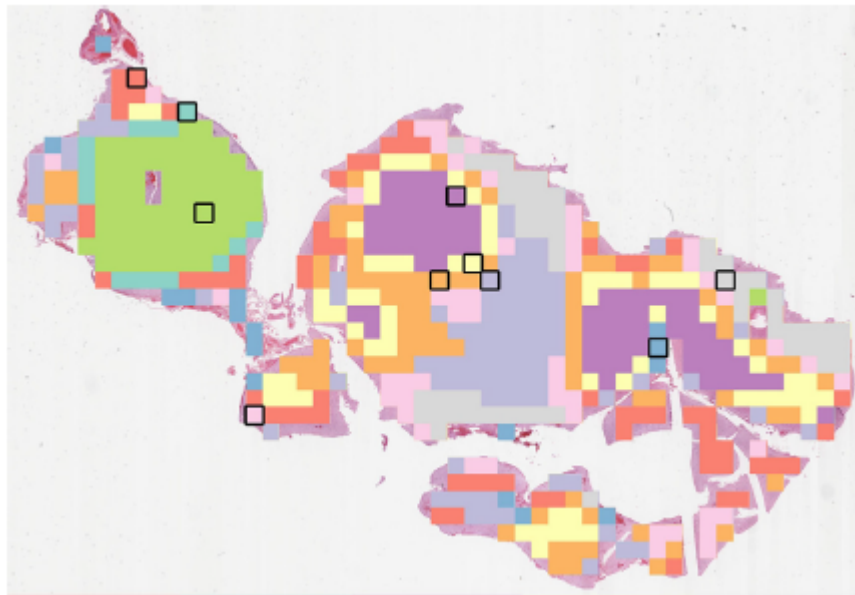
- Coarse and fine feature extraction:



J. Barker *et al.*, "Automated classification of brain tumor type in whole-slide digital pathology images using local representative tiles," *Medical Image Analysis*, 30(1):60-71, 2016.

Clustering-based

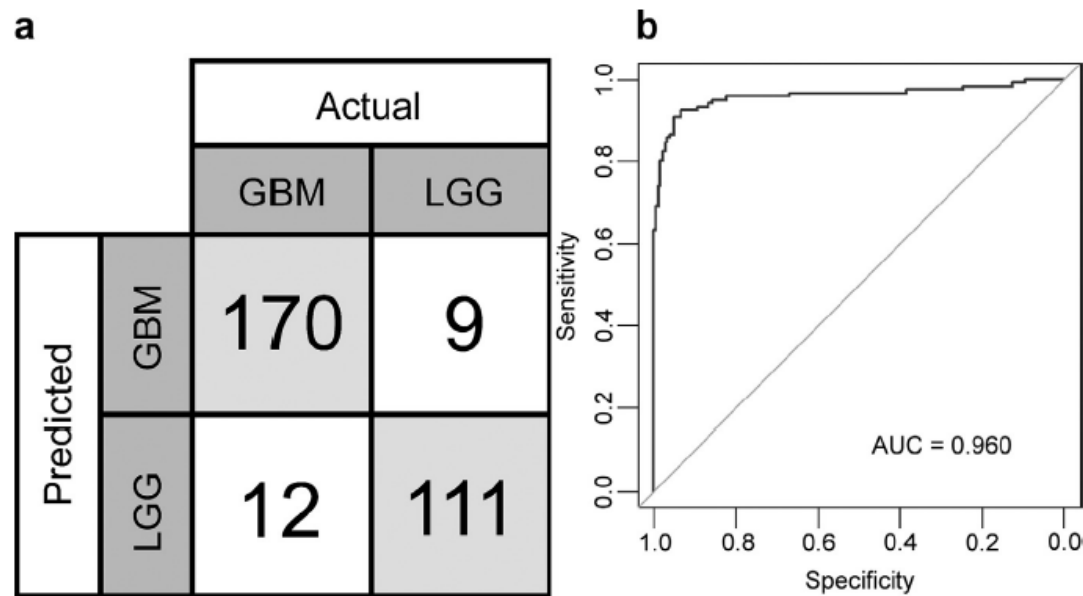
- Coarse and fine feature extraction:
 - Clustering-based representative tile extraction



J. Barker *et al.*, "Automated classification of brain tumor type in whole-slide digital pathology images using local representative tiles," *Medical Image Analysis*, 30(1):60-71, 2016.

Clustering-based

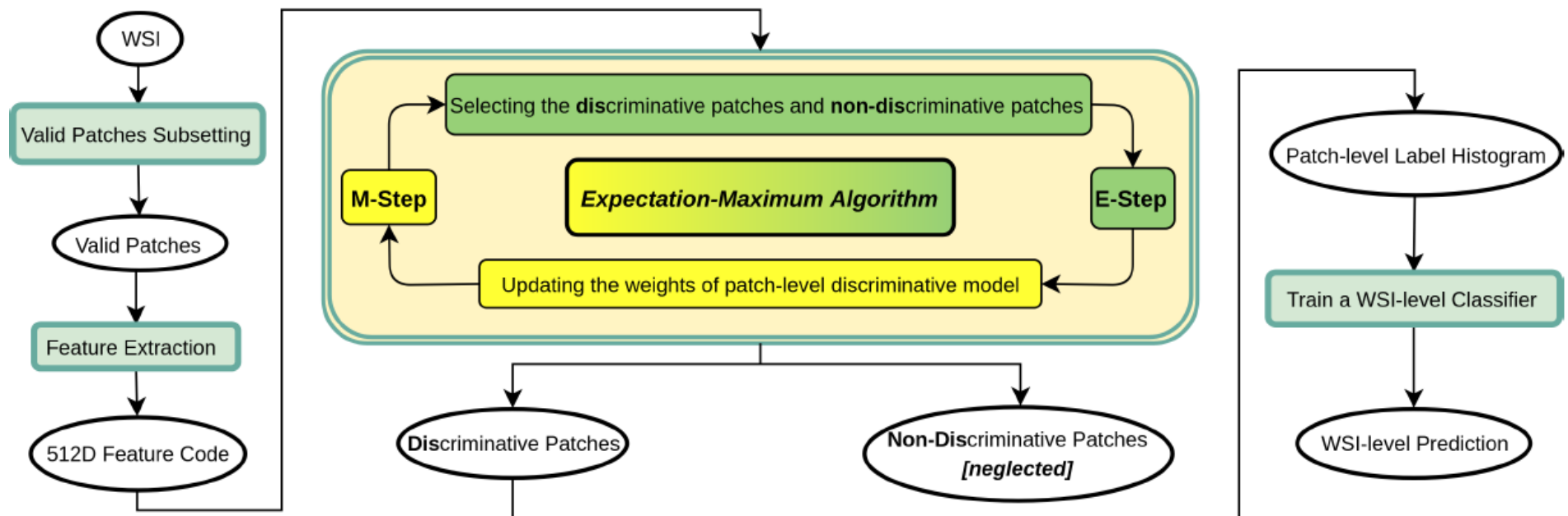
- Coarse and fine feature extraction:



J. Barker *et al.*, "Automated classification of brain tumor type in whole-slide digital pathology images using local representative tiles," *Medical Image Analysis*, 30(1):60-71, 2016.

Pruning-based

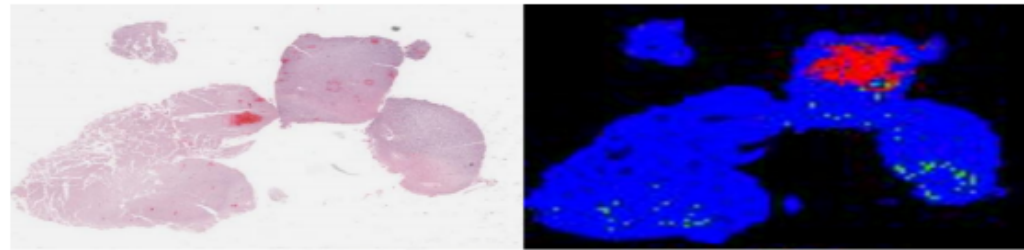
- Discriminative patch-based CNN:
 - EM-based discriminative patch extraction



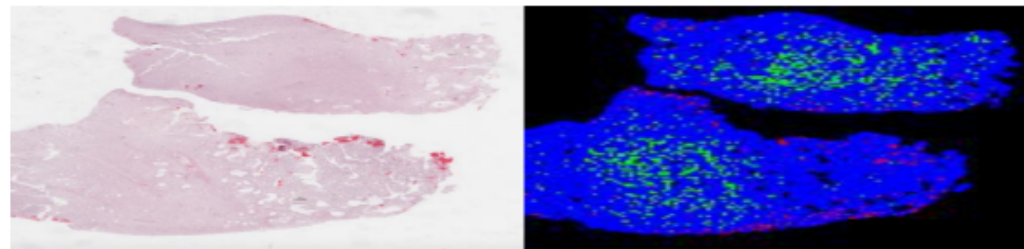
Source: C. Zhang et al. *Whole slide image classification via iterative patch labelling*. ICIP, 2018.

Pruning-based

- Discriminative patch-based CNN:
 - EM-based discriminative patch extraction



(a) Testing oligodendroglioma instance



(b) Testing astrocytoma instance

Source: C. Zhang et al. *Whole slide image classification via iterative patch labelling*. ICIP, 2018.

Pruning-based

- Results on the CBTC challenge dataset:
 - 32 WSI images, with 16 astrocytoma and 16 oligodendroglioma cases

Methods	Acc.
CNN-Feat-SVM	62.50%
Finetune-CNN-Feat-SVM	69.13%
Iter-Finetune-CNN-SVM[Discriminative]	76.62%
Iter-Finetune-CNN-SVM[Both]	84.38%

Source: C. Zhang et al. *Whole slide image classification via iterative patch labelling*. ICIP, 2018.

More Studies

- There are many other approaches for WSI analysis.
 - Do a search in *Google Scholar*
 - Keywords: WSI, microscopy, classification, deep learning

The screenshot shows a Google Scholar search results page. The search bar contains the text "WSI microscopy classification deep learning". The results are sorted by relevance. The first result is "Deep learning for identifying metastatic breast cancer" by D Wang, A Khosla, R Gargeya, H Irshad, et al., published in arXiv preprint arXiv in 2016. The second result is "A survey on deep learning in medical image analysis" by G Litjens, T Kooi, BE Bejnordi, AAA Setio, F Ciompi, et al., published in Medical image analysis in 2017. The third result is "Deep learning in microscopy image analysis: A survey" by F Xing, Y Xie, H Su, F Liu, L Yang, et al., published in IEEE Explore in 2017. The fourth result is "Machine learning methods for histopathological image analysis" by D Komura, S Ishikawa, published in Computational and structural biotechnology journal in 2018. The fifth result is "Deep learning as a tool for increased accuracy and efficiency of histopathological diagnosis" published in Nature in 2017.

← → ↻ 🔒 https://scholar.google.com.au/scholar?hl=en&as_sdt=0%2C5&q=WSI+microscopy+classification+deep+learning&btnG=

Google Scholar WSI microscopy classification deep learning 🔍

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Deep learning for identifying metastatic breast cancer [PDF] arxiv.org
D Wang, A Khosla, R Gargeya, H Irshad... - arXiv preprint arXiv ..., 2016 - arxiv.org
... the primary tool used by pathologists to make diag- noses has been the **microscope** [1]. Limitations of ... interest in developing computa- tional methods to assist in the analysis of **microscopic** im- ages ... post-processing, we aim to identify all cancer lesions within each **WSI** with few ...
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A survey on deep learning in medical image analysis [PDF] arxiv.org
G Litjens, T Kooi, BE Bejnordi, AAA Setio, F Ciompi... - Medical image ..., 2017 - Elsevier
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Deep learning in microscopy image analysis: A survey
F Xing, Y Xie, H Su, F Liu, L Yang - ... **networks** and **learning** ..., 2017 - ieexplore.ieee.org
... The high-dimensional **microscopic** data contain complex patterns and dependences in images such ... analysis in terms of (a) network architectures, (b) tasks, (c) **microscope** types, and ... **microscopy** images, and HeLa cells in dif- ferential interference contrast **microscopy** images [77 ...
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[HTML] **Machine learning methods for histopathological image analysis** [HTML] sciencedirect.com
D Komura, S Ishikawa - Computational and structural biotechnology journal, 2018 - Elsevier
... a human pathologist observing the stained specimen on the slide glass using a **microscope** ... Case level **classification** summarizing patch or object level **classification**, Markov Random Field [17], Bag ... For example, as the entire **WSI** could contain hundreds of thousands of patches ...
☆ ⓘ Cited by 32 Related articles All 9 versions

[HTML] **Deep learning as a tool for increased accuracy and efficiency of histopathological diagnosis** [HTML] nature.com

Summary

- Biomedical imaging is critical in precision medicine.
- There are unique challenges when analysing biomedical images:
 - Insufficient ground truth
 - Highly heterogeneous data
- This is a very active area both in industry and research.